

Session-6

Hedging Using Futures

Long & Short Hedges

A long futures hedge is appropriate when you know you will purchase an asset in the future and want to lock in the price

A short futures hedge is appropriate when you know you will sell an asset in the future and want to lock in the price

Example of Hedging

- Imagine you have bought 250 shares of Infosys at Rs.2,284/- per share. The quarterly results are expected soon. You are worried Infosys may announce a not so favorable set of numbers, as a result of which the stock price may decline considerably. To avoid making a loss in the spot market you decide to hedge the position.

In order to hedge the position in spot, we simply have to enter a counter position in the futures market. Since the position in the spot is 'long', we have to 'short' in the futures market.

Future price = Rs.2,285/-

Lot size= 250

Infosys Price	Long Spot P&L	Short Futures P&L	Net P&L
2200	$2200 - 2284 = -84$	$2285 - 2200 = +85$	$-84 + 85 = +1$
2290	$2290 - 2284 = +6$	$2285 - 2290 = -5$	$+6 - 5 = +1$
2500	$2500 - 2284 = +216$	$2285 - 2500 = -215$	$+216 - 215 = +1$

Example (Exporter)

ADITYA EXPORT has an inward remittance of USD 50,000 after one month. The company has been advised by its bank that the rupee is likely to be strengthened due to heavy inflow of FDI inflows in recent months. To hedge against any rupee appreciation against dollar, the export house is considering using one-month USDINR futures, quoted at USD/INR 74.5675. The current spot rate is USD/INR 74.4577. The lot size of USD-INR futures is 1000 USD.

- (a) What will be the hedging strategy of the exporter?
- (b) What will be the hedging outcome after one month under the following scenarios?

Scenario-1: Rupee appreciates to USD/INR 73.4077 and futures contract price moves to USD/INR 73.5175.

Scenario-2: Rupee depreciates further to USD/INR 75.5040 and futures contract price reaches USD/INR 75.6138.

a) Hedging strategy:

Since the exporter is holding a long position of receivables, it will short (i.e. sell) USDINR futures at USD/INR 74.5675.

The lot size of USD-INR futures is 1000 USD. So, the exporter will short 50 lots, which is computed as follows:

$$\text{Hedge ratio} = (\text{Exposure value} / \text{lot size}) = \underline{50000 / 1000} = \underline{50}$$

b) Hedge outcomes

Loss/gain from <u>spot position</u>	Loss/gain from <u>futures</u>
Scenario-1 (rupee appreciation)	
Cash Loss = $(\underline{74.4577} - \underline{73.5077}) * 50000 = 52500$	Cash Gain = $(74.5675 - \underline{73.5175}) * 50 * 1000 = \underline{52500}$
Scenario-2 (rupee depreciation)	
Cash Gain = $(75.5040 - 74.4577) * 50000 = 52315$	Cash Loss = $(75.6138 - 74.5675) * 50 * 1000 = 52315$

Example (Importer)

SUNIL Enterprise has imported spare parts from Germany and the total value of the shipment is EUR 75,000 payable after 3 months. The current spot rate EUR/INR 80.25. Apprehending that EUR may become weak during the intervening period, the importing is planning to hedge his position with EURINR futures quoting at EUR/INR 80.65

- (a) What will be the hedging strategy of the importer?
- (b) Determine hedging outcome after 3 months if (i) rupee appreciates to EUR/INR 79.50 with futures quoting at 79.90; and (ii) rupee depreciates to EUR/INR 81.75 with futures quoting at 82.15.

(a) Hedging strategy:

Since the importer is holding a short position of receivables, it will long (i.e. buy) EURINR futures at EUR/INR 80.65.

The lot size of EUR-INR futures is 1000 EUR. So, the exporter will long 75 lots, which is computed as follows:

Hedge ratio= (Exposure value/lot size)= 75000/1000=75

b) Hedge outcomes

Loss/gain from cash position	Loss/gain from futures
Scenario-1 (rupee appreciation)	
Cash Gain = $(80.25 - 79.50) \times 75000 = 56250$	Cash Loss = $(80.65 - 79.90) \times 75 \times 1000 = 56250$
Scenario-2 (rupee depreciation)	
Cash Loss = $(81.75 - 80.25) \times 75000 = 112500$	Cash Gain = $(82.15 - 80.65) \times 75 \times 1000 = 112500$

Example

An exporter of castor oil received an export order of 1000 MT on 3 April, delivery due by the end of May (roughly after two month). Due to quality and logistical concerns, she is planning to buy 1000 MT castor seed from the physical market in May, process it into castor oil, and then export. However, she is anticipating significant increase in price in the spot market at the time of purchase in May. The lot size is 10 MT per contract.

- (a) How can she hedge price risk using castor seed futures? The current spot price of castor seed is Rs. 3,825 per 100 kg, and 20 May futures is quoted at Rs. 3,910 per 100 kg.
- (b) What will be the pay-off from the hedge position by the end of May under the following two situations?
 - (i) Spot price and May futures price go up to Rs. 4,150 and Rs. 4,230 respectively
 - (ii) Spot price and May futures price decline to Rs. 3,700 and Rs. 3,780 respectively

The exporter will hedge its position by going long (i.e. buy) 100 castor seed futures contracts.

$$\text{Hedge ratio} = 1,000/10 = 100 \text{ lots}$$

Scenario 1

ST 50

$$\text{Cash loss in spot market} = (4,150 - 3,825) \times 10 \times 1000 = \text{Rs. } \underline{32,50,000}$$

$$\text{Gains from futures closure} = (4,230 - 3,910) \times 10 \times 100 \times 10 = \text{Rs. } \underline{32,00,000}$$

$$\text{Net cash loss} = (32,50,000 - 32,00,000) = \text{Rs. } \underline{50,000}$$

The gains from futures has offset the cash loss to a large extent, but hedging is not perfect. Why? This is because basis has narrowed down from $(3910 - 3825 =) 85$ on 3rd April to 80 by the time of closure of May futures. Thus, whether hedge pay-off would be perfect or not would depend on what happens to basis. This what is called basis risk. **In the process of hedging price risk, we get exposed to basis risk.**

Scenario 2

$$\text{Cash gain in spot market} = (3,825 - 3,700) \times 10 \times 1000 = \text{Rs. } \underline{12,50,000}$$

$$\text{Loss from futures closure} = (3780 - 3910) \times 10 \times 100 \times 10 = (-) \text{Rs. } \underline{13,00,000}$$

The gains from fall in commodity price in may has been eroded by loss from closure of futures.

Basis Risk



BASIS IS USUALLY DEFINED AS THE SPOT
PRICE MINUS THE FUTURES PRICE



BASIS RISK ARISES BECAUSE OF THE
UNCERTAINTY ABOUT THE BASIS WHEN
THE HEDGE IS CLOSED OUT

Short Hedge for Sale of an Asset

Define

F_1 : Futures price at time hedge is set up

F_2 : Futures price at time asset is sold

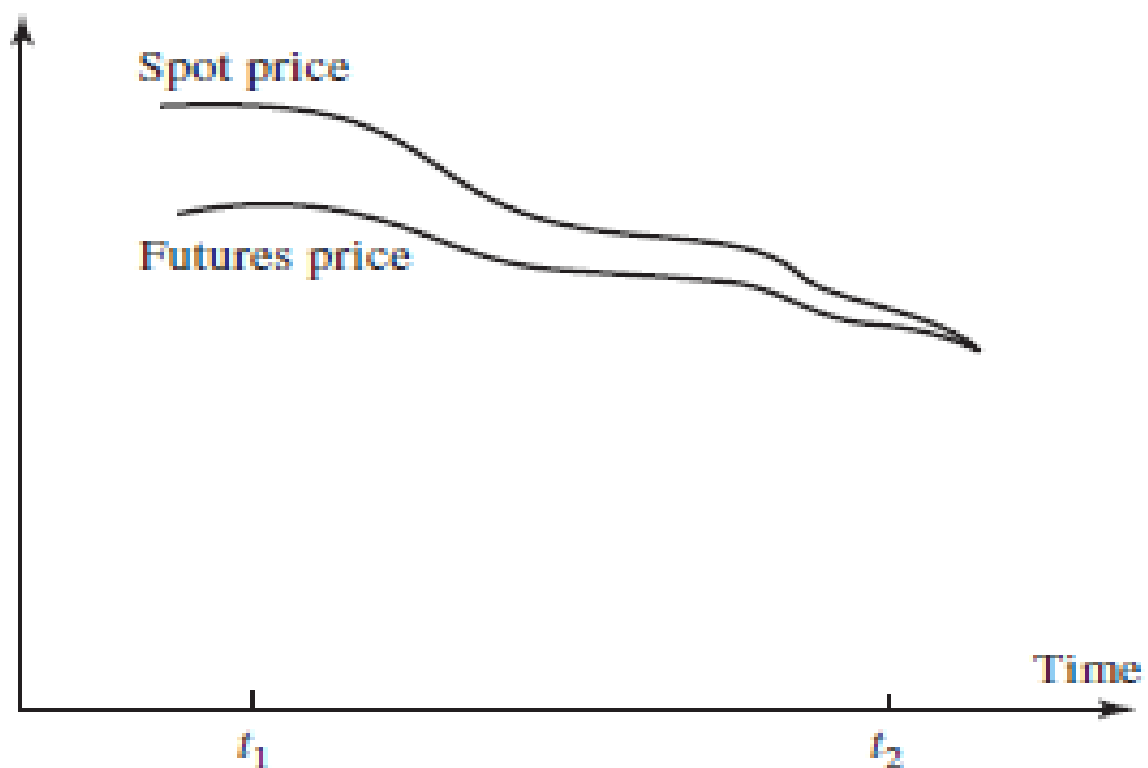
S_1 : Asset price at time of hedge is set up

S_2 : Asset price at time of sale

b_2 : Basis at time of sale

Price of asset	S_2
Gain on Futures	$F_1 - F_2$
Net amount received	$S_2 + (F_1 - F_2) = F_1 + b_2$

Variation of Basis Over Time



Source of Basis

- The assets whose price to be hedged may not be exactly the same as the asset underlying the futures contract
- There may be uncertainty as to the exact date when the assets will be bought or sold
- The hedge may require the futures contract to be closed out before its delivery month

Choice of Contract

- Choose a delivery month that is as close as possible to, but later than, the end of the life of the hedge
- When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price. This is known as cross hedging.

Optimal Hedge Ratio

Proportion of the exposure that should optimally be hedged is

$$h^* = \rho \frac{\sigma_S}{\sigma_F}$$

where

σ_S is the standard deviation of ΔS , the change in the spot price during the hedging period,

σ_F is the standard deviation of ΔF , the change in the futures price during the hedging period

ρ is the coefficient of correlation between ΔS and ΔF .

Example

- Airline will purchase 2 million gallons of jet fuel in one month and hedges using heating oil futures
- Size of the heating oil contract is 42000 gallons
- Find out the hedge ratio and optimal number of future contracts
- From historical data $\sigma_F = 0.0313$, $\sigma_S = 0.0263$, and $\rho = 0.928$

$$h^* = 0.928 \times \frac{0.0263}{0.0313} = 0.78$$

Example continued

- The size of one heating oil contract is 42,000 gallons
- Optimal number of contracts is

$$= 0.78 \times 2,000,000 / 42,000 = 37$$

Optimal Number of Contracts

Q_A Size of position being hedged (units)

Q_F Size of one futures contract (units)

V_A Value of position being hedged (=spot price times Q_A)

V_F Value of one futures contract (=futures price times Q_F)

Optimal number of contracts if
adjustment for daily settlement

$$= \frac{h^* Q_A}{Q_F}$$

Hedging Using Index Futures

To hedge the risk in a portfolio the number of contracts that should be shorted is

$$\beta \frac{V_A}{V_F}$$

where V_A is the value of the portfolio, β is its beta, and V_F is the value of one futures contract

Example

Suppose that a future contract with 4-month maturity is used to hedge the value of a portfolio over the next 3 months in the following situations:

Index level= 1000

Index futures price=1010

Value of the portfolio= 5050000

Risk-free rate= 4% per annum

Dividend yield on index= 1% per annum

Beta of the portfolio= 1.5

Lot size= 250 times of the index

An investor has invested Rs. 95.5 lacs in a portfolio of stocks as shown in the table below. The investor is worried about fall in portfolio value in near futures and intend to hedge the position with Nifty 50 futures currently quoted at 9,025. [Lot size = 75]

Stock	Amount invested (Rs.)	Portfolio weight	Stock beta
ACC	3,00,000	3.14%	1.22
Axis Bank	12,50,000	13.09%	1.40
BPCL	18,00,000	18.85%	1.42
Cipla	6,50,000	6.81%	0.59
DLF	10,00,000	10.47%	1.86
Infosys	7,50,000	7.85%	0.43
L & T	8,50,000	8.90%	1.43
Maruti Suzuki	14,00,000	14.66%	0.95
Reliance	3,00,000	3.14%	1.22
SBI Limited	12,50,000	13.09%	1.40
Total	95,50,000		1.25

Changing Beta

- What position is necessary to reduce the beta of the portfolio to 0.75?
- What position is necessary to increase the beta of the portfolio to 2.0?

Why Hedge Equity Returns

- May want to be out of the market for a while. Hedging avoids the costs of selling and repurchasing the portfolio
- Suppose stocks in your portfolio have an average beta of 1.0, but you feel they have been chosen well and will outperform the market in both good and bad times. Hedging ensures that the return you earn is the risk-free return plus the excess return of your portfolio over the market.

Hedging in India

- Equity market: Futures and Options
- Currency market: Forward and Options
- Interest rate market: Interest Rate Futures, Forward Rate Agreement, Swaps
- Commodity market: Futures

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